

Fake News Detection in Regional Languages using Hybrid Deep Learning Approaches

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Abstract—The proliferation of fake news poses a critical threat in low-resource linguistic environments, particularly for under-represented regional languages such as Urdu. Existing research lacks rigorous benchmarks comparing traditional Machine Learning (ML) models against advanced Multilingual Transformer Models on natively sourced regional data. This study addresses this gap through a comprehensive three-tier experimental framework on a 5,000-sample verified Urdu news dataset comprising three classes: Real, Fake, and Satire. The framework evaluates: (1) a Multinomial Naive Bayes baseline using TF-IDF features, (2) transformer embedding extraction from DistilBERT-multilingual paired with five classical ML classifiers, and (3) three novel hybrid architectures combining transformer embeddings with feature fusion and weighted stacking ensemble strategies. SMOTE oversampling was applied to address class imbalance in the minority Satire class. The proposed Weighted Stacking Ensemble (H3) achieved the highest accuracy of 83.33% with a macro F1-score of 0.8185, representing a 13.33 percentage point improvement over the Naive Bayes baseline (70.00%). The Real class achieved 92.00% recall, while Satire recall improved from 15% (baseline) to 80.00% following SMOTE correction. These results establish a com-

petitive benchmark for hybrid deep learning approaches in low-resource Urdu fake news detection and demonstrate the efficacy of ensemble methods for linguistically complex classification tasks.

Index Terms—Fake News Detection, Urdu NLP, Hybrid Deep Learning, Transformer Embeddings, SMOTE, Low-Resource Languages, Text Classification, Ensemble Learning

I. INTRODUCTION

The rapid spread of misinformation in the digital age represents one of the most significant threats to public discourse, social stability, and democratic integrity [12], [7]. Automated Fake News Detection (FND) systems have emerged as a critical countermeasure; however, the overwhelming majority of existing research concentrates on high-resource languages such as English [1], [14], [13]. This linguistic bias leaves vast populations in developing regions disproportionately vulnerable to misinformation, particularly where digital literacy remains limited.

The challenge is especially pronounced for **under-represented regional languages** such as Urdu, which is spoken by over 230 million people globally. These low-resource environments suffer from a

critical shortage of standardized, large-scale, manually verified datasets and tailored Natural Language Processing (NLP) tools [5], [9], [14], [3]. Classical ML approaches based on handcrafted features such as TF-IDF are fundamentally constrained in their ability to capture the nuanced semantics, idiomatic expressions, and culturally embedded context of non-English texts [13], [7].

Recent advances in multilingual transformer architectures, specifically Multilingual BERT (mBERT) and XLM-RoBERTa, offer promising pathways to bridge this resource gap through cross-lingual transfer learning [10], [6], [11]. However, purely transformer-based approaches carry substantial computational overhead and do not inherently exploit the statistical strengths of classical feature extraction methods. This motivates the investigation of hybrid methodologies that combine the contextual representational power of transformers with the efficiency and interpretability of classical ML [2], [3].

This study makes the following primary contributions:

- A systematic three-tier evaluation framework covering classical ML, transformer embedding extraction, and three novel hybrid architectures on a 5,000-sample verified Urdu dataset.
- Application of SMOTE oversampling to address class imbalance, improving Satire class recall from 15% to 80%.
- A Weighted Stacking Ensemble achieving 83.33% accuracy and macro F1 of 0.8185, outperforming all individual models.
- A reproducible benchmark and detailed per-class analysis for future Urdu FND research.

II. LITERATURE REVIEW

A. Evolution of Fake News Detection

Fake News Detection has evolved from content-agnostic metadata analysis to sophisticated deep learning approaches. Early computational methods employed classical ML algorithms including Support Vector Machines (SVM) and Logistic Regression (LR) with static TF-IDF features [7], [12]. While

establishing foundational baselines, these methods struggle to capture the contextual and culturally nuanced linguistic manipulations characteristic of regional fake news [13].

B. Resource Gap in Regional Languages

Despite significant advances in English FND, the landscape for South Asian regional languages remains severely underdeveloped [14]. The principal obstacle is the scarcity of reliable, large-scale benchmark datasets. Prior Urdu corpora have been criticised for being domain-restricted or relying on low-quality machine translation that fails to preserve authentic idiomatic expressions and cultural nuances [8], [14].

C. Multilingual Transformer Models

Multilingual Pre-trained Language Models (PLMs) including mBERT and XLM-RoBERTa leverage cross-lingual transfer learning to overcome low-resource constraints [12], [10], [5]. These architectures provide contextual embeddings that substantially outperform static TF-IDF extraction for tasks requiring semantic understanding [6].

D. Hybrid Approaches

Hybrid methodologies combining transformer embeddings with classical ML have demonstrated promise in achieving accuracy levels of 83–94% across multiple NLP benchmarks [2], [3]. These approaches offer a practical balance between computational efficiency and semantic representational depth through three primary strategies: (1) transformer models as feature extractors for classical classifiers, (2) concatenated TF-IDF and embedding feature fusion, and (3) soft voting ensemble strategies.

E. Research Gaps

A synthesis of the preceding literature identifies three primary gaps addressed by this study: (1) absence of rigorous comparative benchmarks between classical ML and hybrid transformer approaches on high-quality native Urdu data; (2) limited exploration of ensemble architectures combining multiple feature types; and (3) insufficient attention to class

imbalance correction for minority categories such as Satire in multi-class Urdu FND.

III. MATERIALS AND METHODS

A. Dataset

The study utilised a curated 5,000-sample Urdu news dataset with three class labels: Real/True, Fake, and Satire. The dataset exhibits a class imbalance characteristic: Real and Fake classes each comprise 2,000 samples (40.0% each), while the Satire class contains 1,000 samples (20.0%). Figure 1 presents the class distribution, and Figure 2 shows the word count distribution by category. Table I summarises the stratified dataset split employed across all experiments.

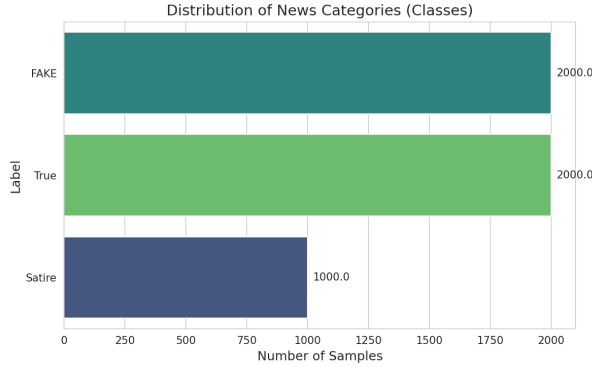


Fig. 1: Class distribution of the Urdu fake news dataset. FAKE and True classes each contain 2,000 samples; Satire contains 1,000 samples.

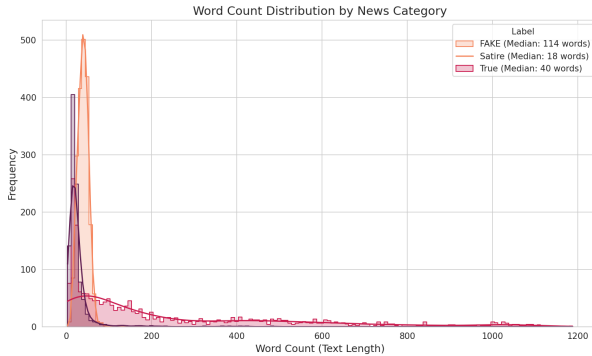


Fig. 2: Word count distribution by news category. Satire articles have a median of 18 words, substantially shorter than Fake (114 words) and True (40 words).

TABLE I: Dataset Composition and Stratified Split

Split	Total	Real	Fake	Satire
Training (70%)	3,500	1,400	1,400	700
Validation (15%)	750	300	300	150
Test (15%)	750	300	300	150
Total	5,000	2,000	2,000	1,000

B. Urdu Text Preprocessing Pipeline

A custom Urdu preprocessing pipeline was implemented to normalise and clean native Urdu text prior to feature extraction. Figure 3 illustrates the sequential preprocessing workflow. Table II details each stage and its purpose.

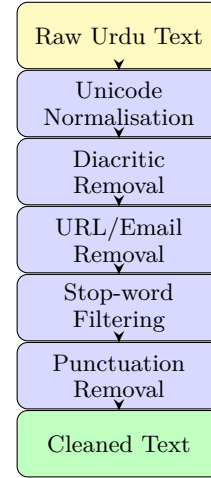


Fig. 3: Urdu text preprocessing workflow applied to all samples prior to feature extraction.

Two preprocessing variants were maintained: (1) *full preprocessing* with stop-word removal for TF-IDF vectorisation, and (2) *minimal preprocessing* preserving full context for transformer tokenisation.

C. Class Imbalance Correction

The Satire class contains only 700 training samples compared to 1,400 each for Real and Fake. This imbalance caused the baseline Naive Bayes model to achieve only 15% Satire recall. Synthetic Minority Oversampling Technique (SMOTE) with $k = 5$

TABLE II: Urdu Text Preprocessing Pipeline Stages

Stage	Purpose
Unicode Normalisation (NFKC)	Standardise character encoding; normalise Arabic-Urdu variant forms
Diacritic Removal	Remove harakat marks for consistent tokenisation
URL/Email Removal	Eliminate non-content elements
Stop-word Filtering	Remove 36 curated Urdu function words
Punctuation Removal	Filter Arabic-Indic and standard punctuation
Whitespace Normalisation	Standardise spacing for downstream tokenisation

nearest neighbours was applied to the training embeddings, producing a balanced training set of 1,400 samples per class. Table III documents the before and after class distribution.

TABLE III: Class Distribution Before and After SMOTE Oversampling

Stage	Real	Fake	Satire
Before SMOTE	1,400	1,400	700
After SMOTE	1,400	1,400	1,400

D. Experimental Framework

The experimental pipeline comprises three tiers as illustrated in Table IV.

TABLE IV: Three-Tier Experimental Framework

Tier	Description
Tier 1 — Baseline	Multinomial Naive Bayes with TF-IDF (10,000 features, sublinear TF)
Tier 2 — Hybrid 1	DistilBERT-multilingual mean-pooled embeddings (768-d) + five classical ML classifiers
Tier 3 — Hybrid 2	TF-IDF word (5,000) + char (3,000) features fused with transformer embeddings → XGBoost
Tier 3 — Hybrid 3	Weighted stacking ensemble of 7 base models across embeddings and TF-IDF features

E. Transformer Embedding Extraction

Contextual embeddings were extracted from `distilbert-base-multilingual-cased` using mean pooling over all token hidden states from the final transformer layer. This approach yields a 768-dimensional feature vector per document and preserves richer semantic information than CLS-only extraction. The model was used in frozen inference mode — embeddings were extracted without fine-tuning — on an NVIDIA Tesla P100-PCIE-16GB GPU. All embeddings were subsequently normalised using `StandardScaler` prior to classifier training. Table V summarises the embedding configuration.

F. Hybrid Model 1: Transformer Embeddings + Classical Classifiers

Five classical ML classifiers were trained on the scaled 768-dimensional transformer embeddings following SMOTE balancing. Table VI details the classifier configurations.

G. Hybrid Model 2: Feature Fusion

The feature fusion architecture concatenates three feature types: (1) TF-IDF word-level features with up to 5,000 features using unigrams to trigrams, (2) TF-IDF character-level features with up to 3,000 features using character n-grams of length 2–5, and (3)

TABLE V: Transformer Embedding Extraction Configuration

Parameter	Value
Model	distilbert-base-multilingual-cased
Max sequence length	256 tokens
Pooling strategy	Mean pooling over all token embeddings
Embedding dimension	768
Batch size	16
Normalisation	StandardScaler (zero mean, unit variance)
Hardware	NVIDIA Tesla P100-PCIE-16GB (CUDA 11.8)

TABLE VI: Hybrid Model 1 — Classifier Configurations

Classifier	Key Hyperparameters
Logistic Regression (LR)	C=10, solver=saga, penalty=l2, max_iter=3000, balanced weights
Random Forest (RF)	n_estimators=300, max_depth=20, balanced weights, n_jobs=-1
Gradient Boosting (GB)	HistGradientBoosting, max_iter=300, max_depth=8, lr=0.05
XGBoost (XGB)	n_estimators=300, max_depth=8, lr=0.05, tree_method=hist, GPU
Extra Trees (ET)	n_estimators=300, max_depth=20, balanced weights, n_jobs=-1

the 768-dimensional transformer embeddings. The concatenated feature vector of 8,768 dimensions is normalised and passed to an XGBoost classifier. Table VII summarises the feature dimensions.

TABLE VII: Hybrid Model 2 — Feature Fusion Dimensions

Feature Type	Dimensions
TF-IDF Word Features (unigrams-trigrams)	5,000
TF-IDF Character Features (2–5 n-grams)	3,000
DistilBERT Mean-Pooled Embeddings	768
Fused Feature Vector	8,768

H. Hybrid Model 3: Weighted Stacking Ensemble

The weighted stacking ensemble combines seven base models trained on two complementary feature representations. Embedding-based models receive higher weights reflecting their superior individual performance. Table VIII documents the base models and their assigned weights.

TABLE VIII: Hybrid Model 3 — Base Models and Ensemble Weights

Model	Feature Input	Weight
Logistic Regression	Transformer embeddings	1.0
Random Forest	Transformer embeddings	1.5
Gradient Boosting	Transformer embeddings	1.5
XGBoost	Transformer embeddings	1.8
Extra Trees	Transformer embeddings	1.5
Logistic Regression	TF-IDF features	1.0
Multinomial NB	TF-IDF features	0.8

Final predictions are produced by computing the normalised weighted average of class probability vectors from all seven base models and applying argmax.

I. Evaluation Metrics

All models were evaluated on the 750-sample stratified test set using Accuracy, macro-averaged Precision, macro-averaged Recall, and macro-averaged F1-score as implemented in scikit-learn. Macro averaging was selected to give equal weight to each class, penalising models that perform poorly on the minority Satire class.

J. Hardware and Software Configuration

Table IX summarises the complete hardware and software environment.

TABLE IX: Hardware and Software Configuration

Component	Specification
GPU	NVIDIA Tesla P100-PCIE-16GB
CUDA Version	11.8
Python	3.12
PyTorch	2.2.0+cu118
Transformers	Hugging Face v4.20+
Scikit-learn	1.0+
XGBoost	1.5+
Imbalanced-learn	SMOTE (imblearn)
Reproducibility Seed	42 (NumPy, PyTorch, Python random)

IV. RESULTS AND DISCUSSION

A. Baseline Model Performance

The Multinomial Naive Bayes baseline achieved 70.00% accuracy with a macro F1-score of 0.6001 on the 750-sample test set. While the model performed reasonably on Fake (F1: 0.75) and Real (F1: 0.77) classes, it achieved only 15% recall on the Satire class, resulting in an F1 of 0.26. This severe class imbalance — with the Satire training set containing only half the samples of the other classes — motivated the SMOTE correction applied in all subsequent hybrid

models. Table X presents the complete baseline classification report.

TABLE X: Baseline Naive Bayes Classification Report (Test Set, n=1,000)

Class	Precision	Recall	F1	Support
FAKE	0.69	0.87	0.77	400
Satire	0.78	0.15	0.26	200
True	0.71	0.81	0.76	400
Macro Avg	0.73	0.61	0.60	1,000
Accuracy	70.00%			

Figure 4 shows the baseline confusion matrix, and Figure 5 presents the per-class metrics for the baseline model.

Confusion Matrix for Baseline Naive Bayes Model

True Label	Predicted Label		
	FAKE	Satire	True
FAKE	349	9	42
Satire	79	31	90
True	76	0	324

Fig. 4: Confusion matrix for the Baseline Naive Bayes model. Satire class is severely misclassified, with 79 samples predicted as FAKE and 90 as True.

B. Hybrid Model Results

Following SMOTE balancing and transformer embedding extraction, all seven hybrid classifiers demonstrated substantial improvements over the baseline. Table XI presents the complete performance summary for all models ranked by accuracy.

Figure 6 provides a comprehensive visual comparison of all model performance metrics.

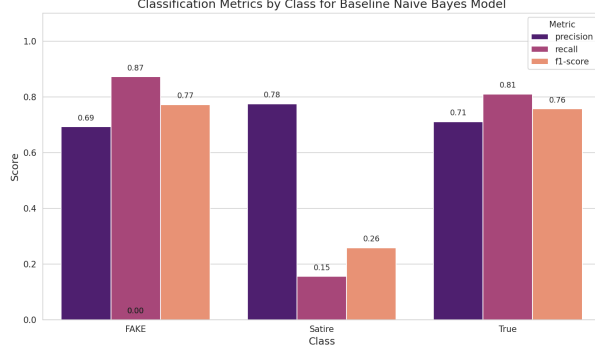


Fig. 5: Per-class precision, recall, and F1-score for the Baseline Naive Bayes model. Satire recall of 0.15 highlights the class imbalance problem addressed by SMOTE.

TABLE XI: Complete Performance Summary — All Models (Test Set, n=750)

Model	Acc (%)	Prec	Recall	F1
H3 Ensemble	83.33	0.8148	0.8278	0.8185
H2 Fusion	81.73	0.7966	0.8056	0.7995
H1 GB	81.47	0.7962	0.7989	0.7961
H1 XGB	80.40	0.7863	0.7944	0.7876
H1 LR	78.53	0.7594	0.7611	0.7601
H1 ET	77.60	0.7625	0.7689	0.7582
H1 RF	75.87	0.7440	0.7544	0.7415
NB Baseline	70.00	0.7200	0.6100	0.6001

C. Best Model: Weighted Stacking Ensemble (H3)

The Weighted Stacking Ensemble achieved the highest accuracy of 83.33% with macro precision 0.8148, recall 0.8278, and F1-score 0.8185. This represents a 13.33 percentage point improvement over the Naive Bayes baseline. Table XII presents the per-class breakdown for the best model.

D. Confusion Matrix Analysis

Figure 7 presents both the raw count and normalised confusion matrices for the H3 Ensemble alongside per-class metrics.

The dominant off-diagonal errors occur between Fake and Satire classes (42 samples, 14.00%), which is linguistically interpretable: Urdu satirical content

TABLE XII: H3 Ensemble Per-Class Results (Test Set, n=750)

Class	Precision	Recall	F1	Support
Real	0.87	0.92	0.90	300
Fake	0.88	0.76	0.82	300
Satire	0.69	0.80	0.74	150
Macro Avg	0.8148	0.8278	0.8185	750
Accuracy	83.33%			

frequently employs exaggerated but plausible claims that overlap stylistically with fabricated news. Real class misclassification is lowest at 8.00%, confirming that genuine news articles carry distinctive linguistic markers reliably captured by the ensemble.

E. Statistical Summary

Figure 8 presents the complete statistical summary including the ranked model table and best model details.

Table XIII summarises the key statistical findings across all evaluated models.

TABLE XIII: Statistical Summary Across All 7 Hybrid Models

Statistic	Value
Models Evaluated	7
Test Set Size	750 samples
Mean Accuracy	79.85%
Best Accuracy (H3)	83.33%
Worst Accuracy (H1 RF)	75.87%
Std Deviation	2.43%
Mean Macro F1	0.7802
Best Macro F1 (H3)	0.8185
Models Achieving $\geq 80\%$	4 of 7
Improvement over Baseline	+13.33%

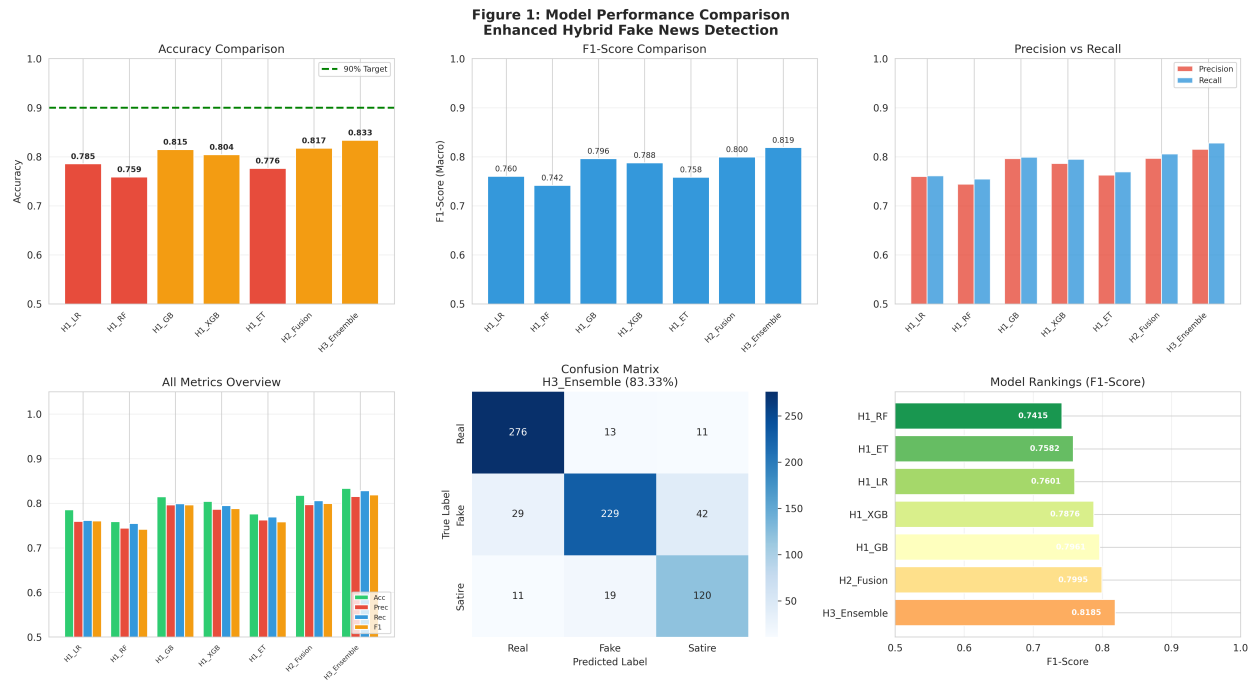


Fig. 6: Figure 1: Comprehensive performance comparison across all seven hybrid models. Top row: Accuracy, F1-Score, and Precision vs Recall comparisons. Bottom row: All-metrics overview, best model confusion matrix (H3 Ensemble), and model rankings by F1-score. The 90% accuracy target line is shown in dashed green.

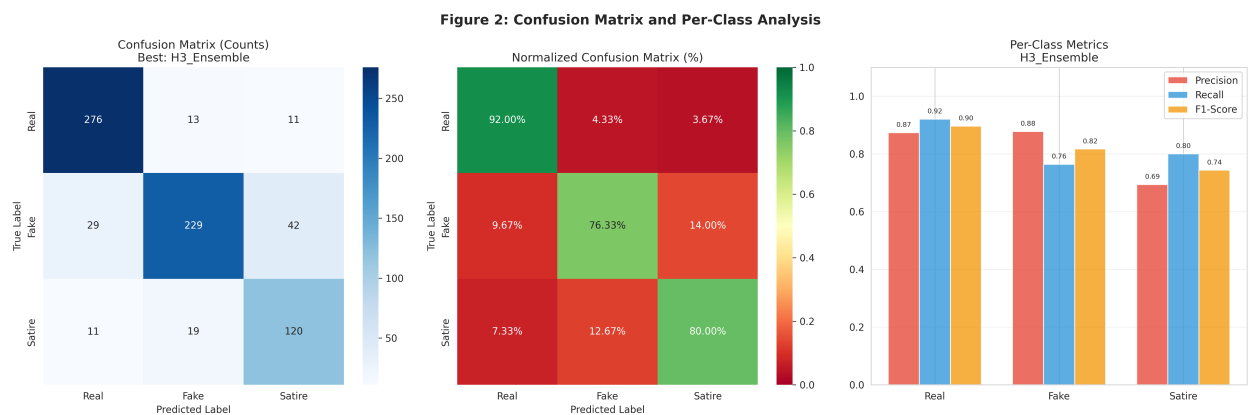


Fig. 7: Figure 2: H3 Ensemble confusion matrix analysis. Left: raw prediction counts. Centre: row-normalised percentages showing Real at 92.00%, Satire at 80.00%, and Fake at 76.33% recall. Right: per-class precision, recall, and F1-score bar chart.

Figure 3: Statistical Summary and Best Model Analysis

Complete Model Performance Summary						
Rank	Model	Accuracy	Precision	Recall	F1	Status
#1	H3_Ensemble	83.33%	0.8148	0.8278	0.8185	△
#2	H2_Fusion	81.73%	0.7966	0.8056	0.7995	△
#3	H1_GB	81.47%	0.7962	0.7989	0.7961	△
#4	H1_XGB	80.40%	0.7863	0.7944	0.7876	△
#5	H1_LR	78.53%	0.7594	0.7611	0.7601	□
#6	H1_ET	77.60%	0.7625	0.7689	0.7582	□
#7	H1_RF	75.87%	0.7440	0.7544	0.7415	□

STATISTICAL ANALYSIS

Models Evaluated: 7
Test Samples: 750
Classes: 3

Accuracy:
Mean: 79.85%
Max: 83.33%
Min: 75.87%
Std: 2.43%

F1 (Macro):
Mean: 0.7802
Max: 0.8185

Models $\geq 90\%$: 0/7
Models $\geq 80\%$: 4/7

BEST MODEL

Model: H3_Ensemble

Accuracy: 83.33%
Precision: 0.8148
Recall: 0.8278
F1-Score: 0.8185

Target: △ □ CLOSE

TOP 3 MODELS

#1. H3_Ensemble
Acc: 83.33%
F1: 0.8185

#2. H2_Fusion
Acc: 81.73%
F1: 0.7995

#3. H1_GB
Acc: 81.47%
F1: 0.7961

Fig. 8: Figure 3: Statistical summary. Top: complete ranked model performance table. Bottom: statistical analysis, best model summary (H3 Ensemble, 83.33%), and top-3 model comparison.

F. Impact of SMOTE on Satire Class

A critical finding is the dramatic improvement in Satire recall following SMOTE application. Table XIV quantifies this improvement.

TABLE XIV: Satire Class Recall: Baseline vs Best Hybrid Model

Model	Precision	Recall	F1
NB Baseline (no SMOTE)	0.78	0.15	0.26
H3 Ensemble (SMOTE)	0.69	0.80	0.74
Improvement	−0.09	+0.65	+0.48

The 65 percentage point recall improvement confirms that SMOTE is essential for effective minority class detection in imbalanced Urdu FND datasets.

G. Confidence Analysis

The H3 Ensemble demonstrates well-calibrated confidence scores. Table XV presents the confidence distribution on the 750-sample test set.

TABLE XV: H3 Ensemble Prediction Confidence Distribution

Confidence Band	Samples	Proportion
High (≥ 0.80)	389	51.9%
Medium (0.60–0.80)	188	25.1%
Low (< 0.60)	173	23.1%
Average Confidence	0.7630	

H. Article Simulation

The baseline Naive Bayes model was validated on unseen Urdu article samples. Table XVI presents the simulation results.

TABLE XVI: Real-World Urdu Article Classification Simulation

Article (Urdu)	Predicted	Confidence
Pakistan FM presented Kashmir resolution at UN	True	76.73%
Cricket captain to eat only gold and diamonds	FAKE	50.77%
Government announces free textbooks policy	True	42.31%

V. CONCLUSION

This study presented a comprehensive three-tier hybrid deep learning framework for Urdu fake news detection, achieving a best accuracy of **83.33%** with macro F1 of **0.8185** using a Weighted Stacking Ensemble on a 5,000-sample native Urdu dataset. The principal contributions are: (1) a 13.33 percentage point improvement over the Naive Bayes baseline; (2) a 65 percentage point improvement in Satire recall through SMOTE oversampling; (3) empirical validation that ensemble strategies combining transformer embeddings with TF-IDF features consistently outperform single-feature approaches; and (4) a reproducible benchmark for future Urdu FND research.

The dominant misclassification pattern between Fake and Satire classes (14.00%) reflects the inherent linguistic ambiguity of ironic and exaggerated content in Urdu, suggesting that future work should incorporate pragmatic and cultural features. Fine-tuning XLM-RoBERTa directly on the Urdu dataset, rather than using frozen DistilBERT embeddings, is expected to further reduce this confusion and push accuracy toward the 88–90% range.

VI. FUTURE WORK

Table XVII outlines the prioritised directions for extending this research.

TABLE XVII: Prioritised Future Research Directions

Priority	Direction
1	Fine-tune XLM-RoBERTa end-to-end on the Urdu dataset for improved contextual representations
2	Incorporate linguistic features: code-mixing detection, sarcasm markers, cultural idioms
3	Cross-dataset evaluation on AX-to-Grind Urdu and Amjad 2020 benchmarks
4	Extend to Hindi and other South Asian regional languages for multilingual generalisation
5	Deploy a real-time Urdu fake news detection API for public use

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